



FOR OFFICIAL USE

--	--	--	--	--	--

National
Qualifications
2024

Mark

--

X823/75/01**Engineering Science**

FRIDAY, 17 MAY

9:00 AM – 10:50 AM



* X 8 2 3 7 5 0 1 *

Fill in these boxes and read what is printed below.

Full name of centre

--

Town

--

Forename(s)

--

Surname

--

Number of seat

--

Date of birth

Day

--	--

Month

--	--

Year

--	--

Scottish candidate number

--	--	--	--	--	--	--	--	--	--

Total marks — 110**SECTION 1 — 20 marks**

Attempt ALL questions.

SECTION 2 — 90 marks

Attempt ALL questions.

Show all working and units where appropriate.

You should refer to the National 4/5 Engineering Science Data Booklet which you have been given.

The number of significant figures expressed in a final answer should be equivalent to the least significant data value given in the question. Answers that have two more figures or one less figure than this will be accepted.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



* X 8 2 3 7 5 0 1 0 1 *

SECTION 1 — 20 marks

Attempt ALL questions

1. Digital electronic signals can be controlled using logic gates.

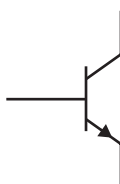
- (a) State the logic gate with a **single** input that will change a high input to a low output.

1

- (b) Draw the symbol for an AND gate.

1

2. The symbol for an electronic component is shown below.



- (a) State the name of this component.

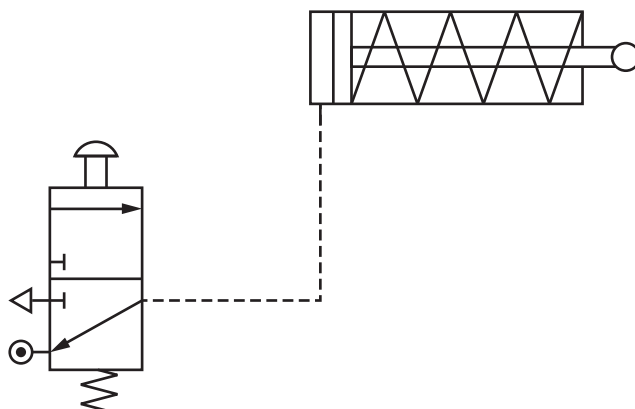
1

- (b) Indicate with an X, the location of the **base** connection on the symbol shown above.

1



3. A diagram for a pneumatic circuit, designed to outstroke a piston when a push button is pressed on a valve, is shown.



Errors were made in the circuit diagram shown above.

- (a) Describe two errors that were made in this circuit diagram.

2

Error 1 _____

Error 2 _____

- (b) State the type of engineer that would be responsible for designing a pneumatic circuit.

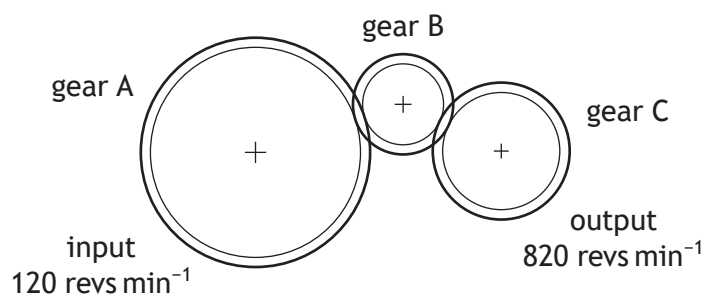
1

[Turn over



* X 8 2 3 7 5 0 1 0 3 *

4. A diagram of a simple gear train is shown below.



- (a) Calculate the velocity ratio of this simple gear train.

2

- (b) State the effect on the output velocity if the number of teeth on gear B is **reduced**.

1



5. A wind farm is being developed on a Scottish island.



- (a) State the type of engineer that would be responsible for monitoring:

(i) the construction of the access roads to the wind turbines

1

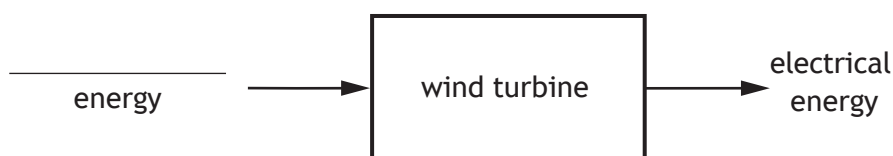
(ii) the impact on the land during installation of the wind turbines.

1

An incomplete diagram showing the **energy** transformation in a wind turbine is shown below.

- (b) Complete the diagram by adding the missing input **energy**.

1



The wind turbine uses feedback in its operation.

- (c) State the type of control that uses feedback.

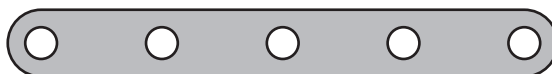
1



6. Members in a frame structure are tested using compressive and tensile forces.

- (a) (i) Draw two arrows which represent an external **compressive** force on member A shown below.

1



member A

- (ii) Draw two arrows which represent an external **tensile** force on member B shown below.

1



member B

Member B has a cross-sectional area of 36 mm^2 .

- (b) Calculate the stress on member B when a 1152 N force is applied.

2



7. Friction can cause the moving parts of a drive system to wear out.

(a) Describe how to reduce the wear on the moving parts of a drive system.

1

(b) State a form of energy lost due to friction in a drive system.

1

[Turn over



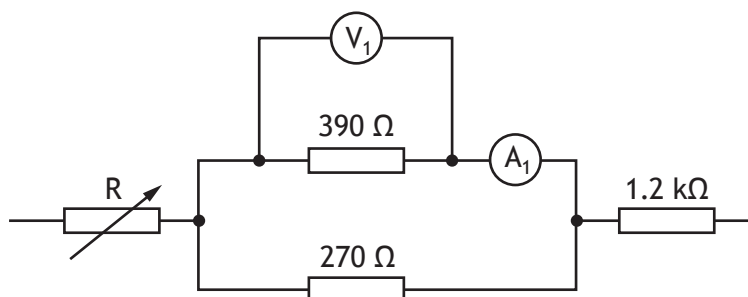
SECTION 2 — 90 marks

Attempt ALL questions

8. A child's microphone with built-in speaker is shown.



Part of the circuit used to control the volume of the speaker is shown.



- (a) For the circuit shown above:

- (i) calculate the resistance of the **parallel** branch

2



8. (a) (continued)

- (ii) calculate the **total resistance** of the circuit when the variable resistor R is set to $780\ \Omega$.

2

- (b) Calculate the current reading on ammeter A_1 when the voltage V_1 is 1.2 V.

3

The resistance of the variable resistor R in the circuit is **increased**.

- (c) State the effect this will have on the:

(i) voltage V_1 _____

1

(ii) current A_1 _____

1

During the development of the child's microphone and speaker, computer simulation was used to test the complete circuit before constructing a prototype.

- (d) Explain why it is an advantage to test the circuit using computer simulation before constructing the prototype.

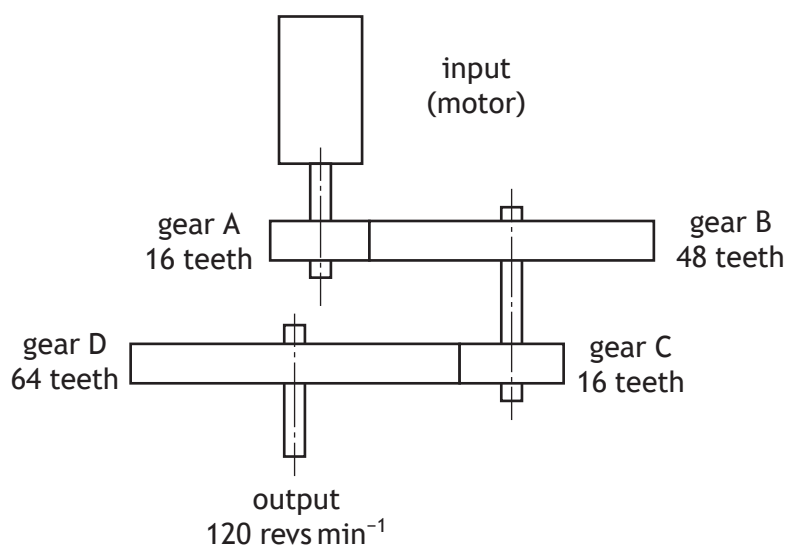
2



9. An electric bike uses a motor to assist a cyclist when pedalling. An electric bike and charging point are shown.



The motor and gear train used in the electric bike are shown below.



9. (continued)

(a) Calculate the rotational speed of the motor.

4

The motor is rated at 36 V and 250 W.

(b) Calculate the current supplied to the motor.

3



9. (continued)

(c) Explain how the efficiency of the gear train could be improved.

2

(d) Describe one positive **environmental** impact of using an electric bike.

1

(e) Describe one positive **social** impact of using an electric bike.

1

(f) Describe one negative **economic** impact of using an electric bike.

1



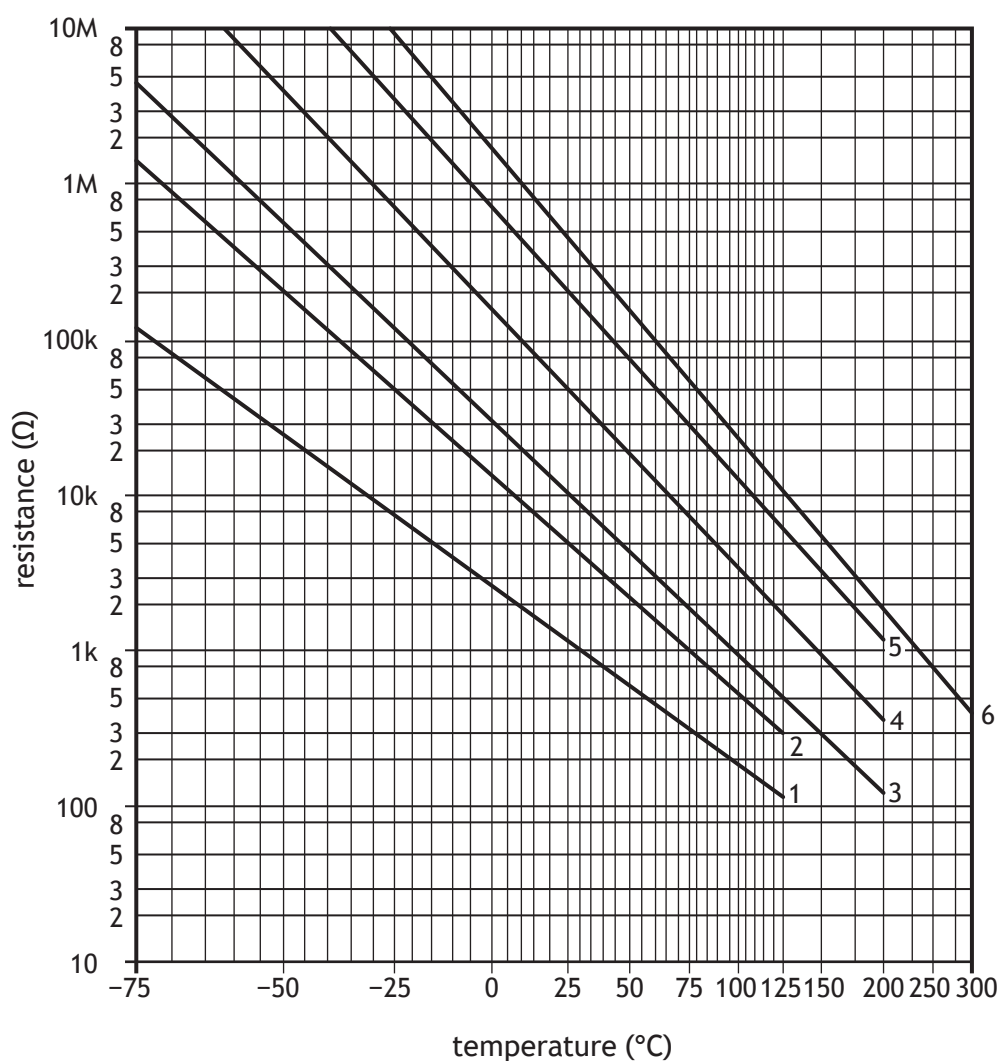
10. A kettle is shown below.

MARKS
DO NOT
WRITE IN
THIS
MARGIN



A thermistor is used as part of a sensor to measure the water temperature in the kettle.

The operating characteristics for a range of thermistors are shown on the graph below.



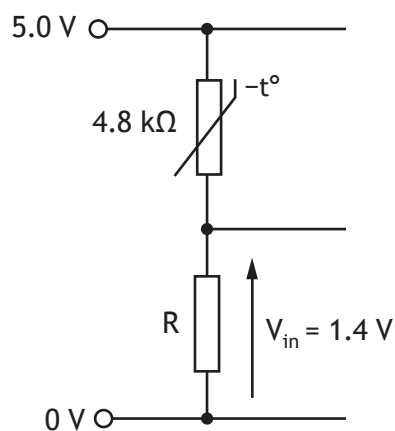
- (a) Determine the resistance of a **type 2** thermistor at a temperature of 85 °C, with reference to the graph.

1



10. (continued)

The sensing circuit used to detect the temperature of the water is shown below.



(b) State the name of this type of series circuit.

1

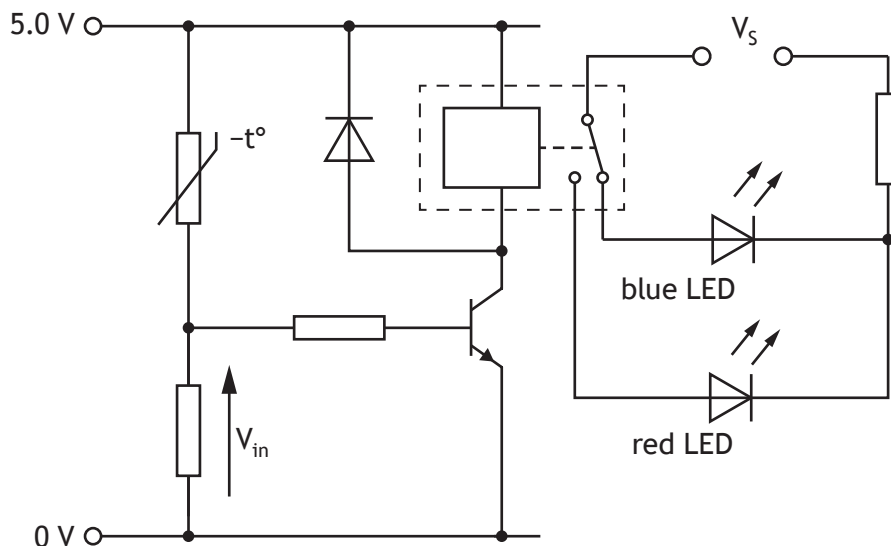
(c) Calculate the resistance R in the sensing circuit shown above.

4



10. (continued)

The kettle has coloured LEDs to indicate the temperature of the water. A simplified circuit used to model the operation of the LEDs is shown below.



- (d) Describe the operation of this circuit as the water temperature increases. Include reference to the resistance of the thermistor, the voltage V_{in} and both LEDs.

5

As the water temperature increases . . . _____

[illegible]

10. (continued)

A kettle uses established technology.

An **emerging technology** is one that is being developed and still to be tried commercially within a product or system.

- (e) Explain a possible impact of an emerging technology which you are familiar with.

2

Name of emerging technology _____

Impact _____



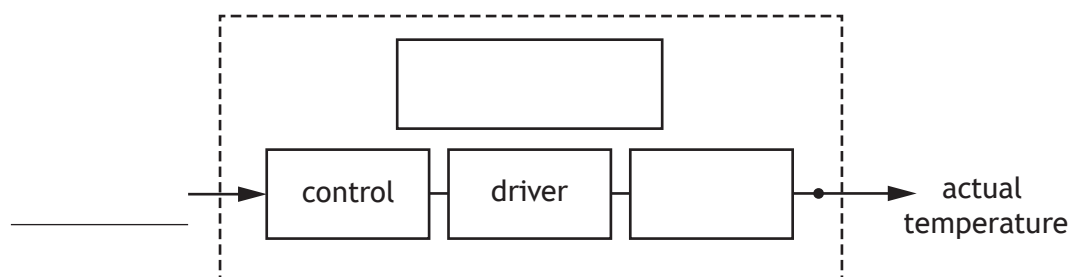
11. A dishwasher in a restaurant kitchen is shown.



Part of the specification for the dishwasher is given below:

- A user selects a set temperature.
- A sensor detects the actual water temperature and sends a signal to the control.
- The control compares the actual water temperature with the set temperature.
- If the actual water temperature is lower than the set temperature a heating element will switch on.

An incomplete sub-system diagram for the dishwasher is shown below.



(a) Complete the sub-system diagram with reference to the specification.

4

[Turn over



11. (continued)

The dishwasher is operated by a microcontroller.

Input and output connections to the microcontroller are shown in the table below.

Input connection	Pin	Output connection
	7	red LED
	6	water pump
	5	heating element
	4	buzzer
	3	
start switch	2	
door sensor	1	
	0	

A drying cycle used in the dishwasher operates using the following sequence:

- When a start switch is pressed a red LED will turn on.
- After a door sensor is activated a water pump will turn on for 35 seconds and then turn off.
- A heating element will then turn on for 5 minutes and then turn off.
- A buzzer will then sound six times, turning on for 0.3 seconds and turning off for 0.3 seconds each time.
- The red LED will then turn off.
- The sequence will return to the start.

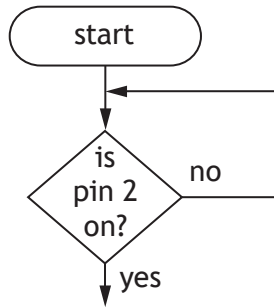
(b) Complete the flowchart shown opposite for the sequence with reference to the data booklet and input/output connections.

Include **all** pin numbers and delay units in your flowchart.

11



11. (b) (continued)



[Turn over



12. A portable power supply used when camping can produce a range of output voltages.

The portable power supply can be charged using solar panels as shown below.



- (a) Explain why using solar panels can contribute towards limiting climate change. 2

A 24 V mini fridge is powered by the portable power supply for 7.5 hours.

- (b) Calculate the electrical energy used by the mini fridge when the current is 0.42 A. 3



12. (continued)

The Boolean equation for part of a circuit used in the portable power supply is shown.

$$Z = (A + B) \bullet (\bar{A} + C)$$

(c) Draw the logic diagram for this Boolean equation.

4

A ○

B ○

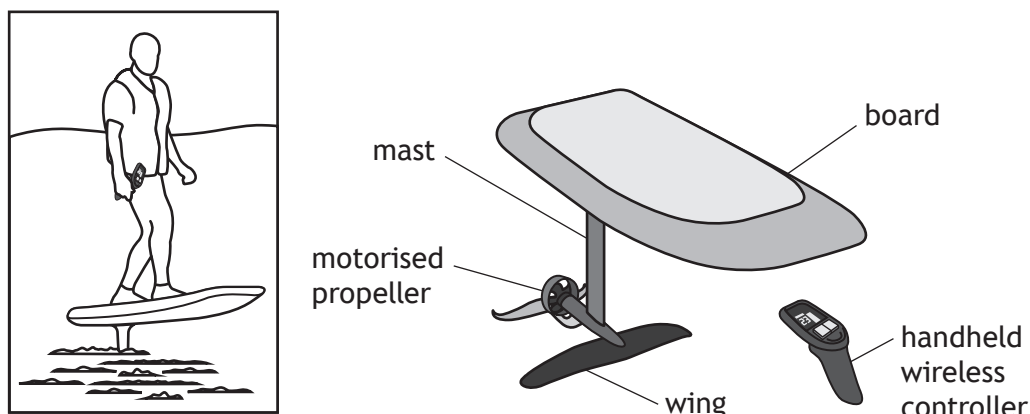
○ Z

C ○

[Turn over



13. A hydrofoil board allows a rider to glide over water.
The hydrofoil board and handheld wireless controller are shown.



Several different engineers were involved during their development.

- (a) Describe a task that each of the following types of engineer would complete during the **development** of the hydrofoil board and handheld wireless controller:

(i) mechanical engineer

1

(ii) structural engineer

1

(iii) electronic engineer.

1



13. (continued)

The hydrofoil board and rider have a combined mass of 115 kg.

- (b) Calculate the velocity of the hydrofoil board and the rider when their kinetic energy is 13 kJ.

3

The hydrofoil board is to be used in the sea and must be able to withstand a range of different forces.

The properties of four materials considered for the mast are shown in the table below.

Material	Corrosion resistant	Strength
A	yes	low
B	no	high
C	yes	high
D	yes	medium

- (c) Select, from the table, the most suitable material to be used for the mast and justify your choice.

2

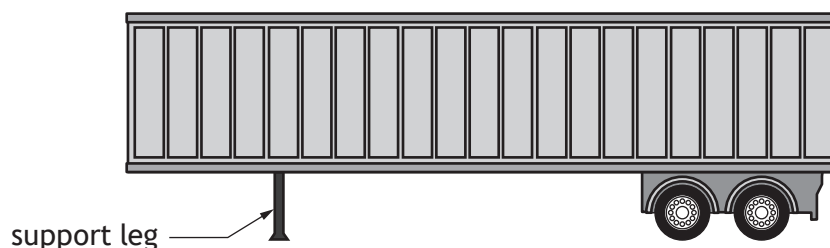
Selected material _____

Justification _____

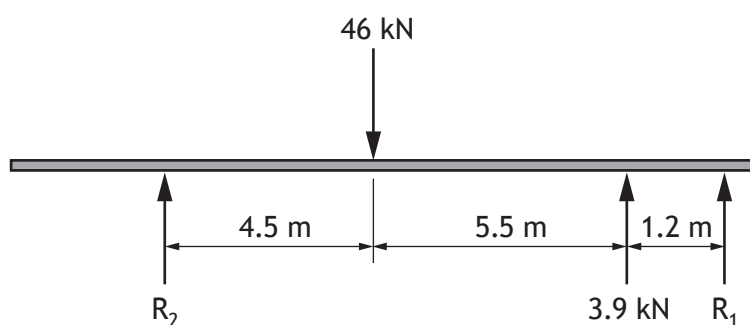
[Turn over



14. A refrigerated lorry trailer used to transport fresh food is shown below.



A simplified free body diagram for the empty lorry trailer is shown.



- (a) (i) Calculate the reaction force R_1 by taking moments about R_2 .

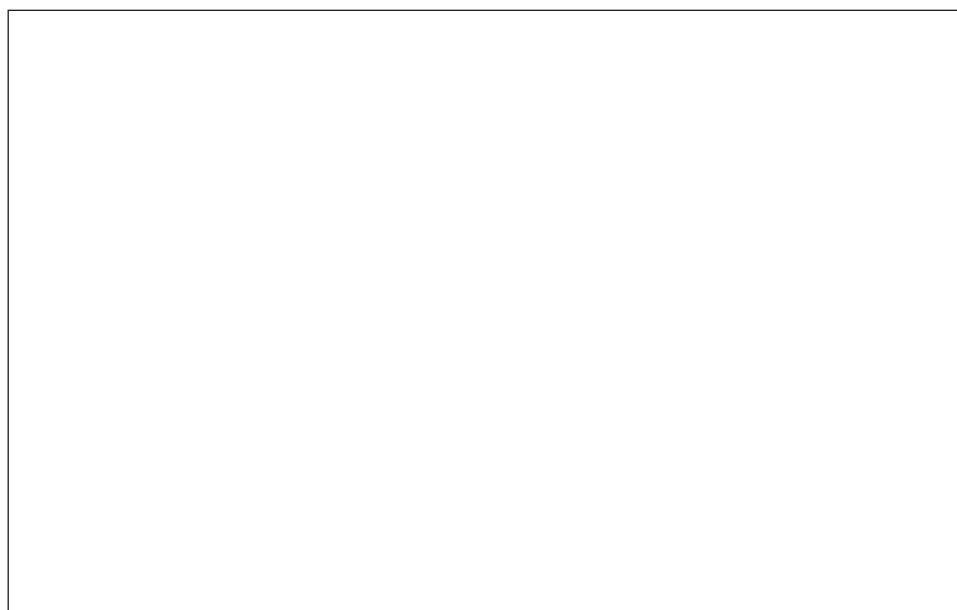
3



14. (a) (continued)

(ii) Calculate the reaction force R_2 .

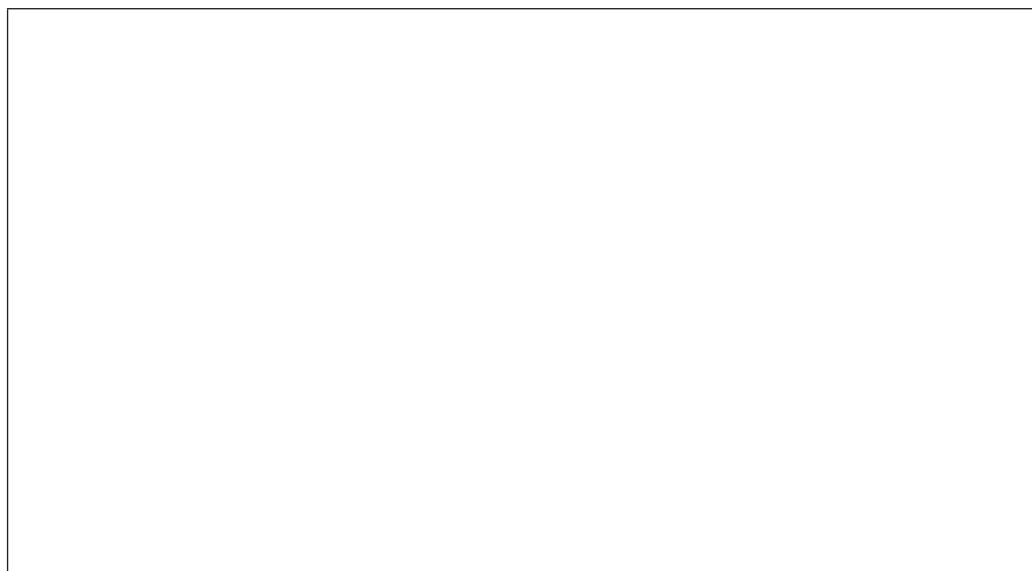
2



When loaded, the lorry trailer's support leg has a strain of 0.00038 and a change in length of 0.46 mm.

(b) Calculate the original length of the support leg.

3



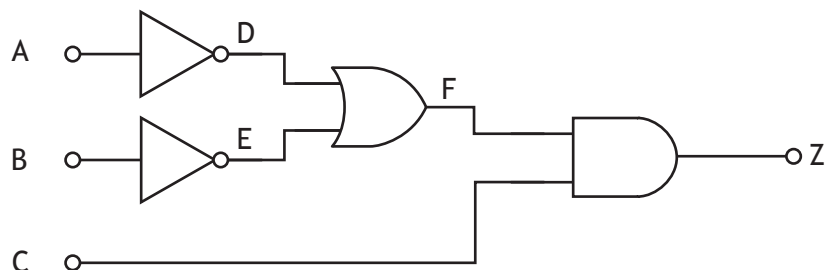
[Turn over



14. (continued)

The temperature inside the trailer is monitored.

Part of the logic circuit used in the temperature monitoring system is shown below.



(c) Complete the truth table for the logic circuit shown.

3

A	B	C	D	E	F	Z
0	0	0	1			
0	0	1	1			
0	1	0	1			
0	1	1	1			
1	0	0	0			
1	0	1	0			
1	1	0	0			
1	1	1	0			

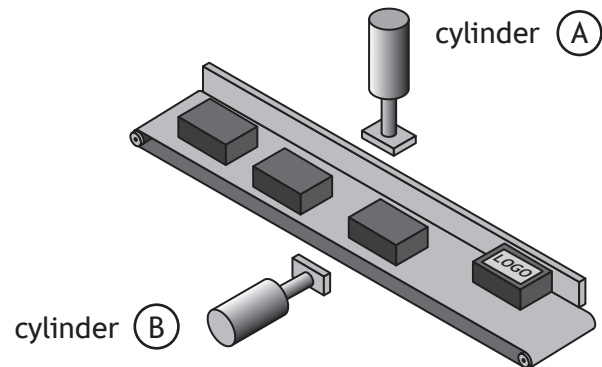


[Turn over for next question

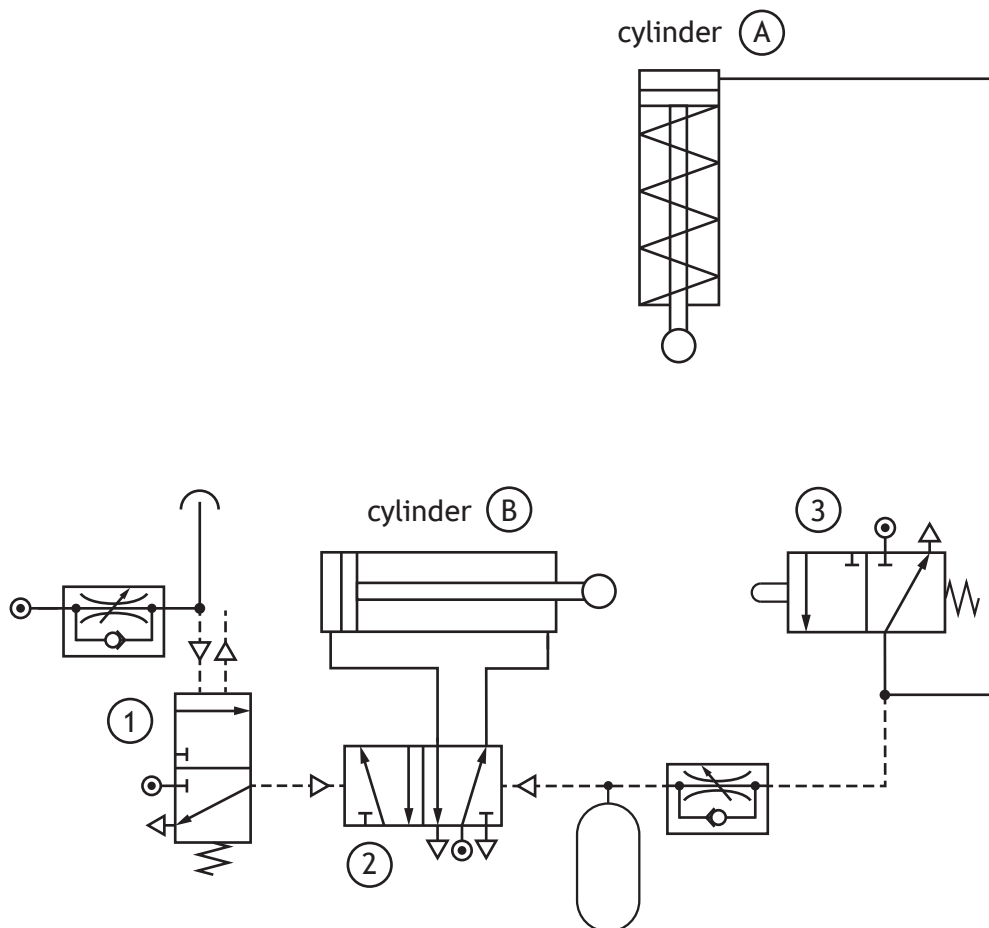
DO NOT WRITE ON THIS PAGE



15. Pneumatics is used to stamp a logo onto products.



A diagram for a pneumatic circuit that is used to stamp the logo onto the products is shown below.



* X 8 2 3 7 5 0 1 2 8 *

15. (continued)

- 5

When the air bleed is covered valve (1) is actuated . . . _____

[illegible]

Cylinder (B) requires an outstroking force of 18 N.

- 3

Case No.	Case Name	Case Address	Case City	Case State	Case Zip	Case Phone	Case Email	Case Date	Case Time	Case Status	Case Notes
1	John Doe	123 Main St	New York	NY	10001	212-555-1234	john.doe@example.com	2023-10-27	14:30	Completed	Initial assessment and data collection.
2	Jane Smith	456 Elm St	Los Angeles	CA	90001	310-555-5678	jane.smith@example.com	2023-10-28	10:00	In Progress	Interview with subject and review of records.
3	Robert Johnson	789 Oak St	Chicago	IL	60601	312-555-9012	robert.johnson@example.com	2023-10-29	09:00	Pending	Awaiting further information from the subject.
4	Maria Garcia	101 Pine St	San Francisco	CA	94101	415-555-3456	maria.garcia@example.com	2023-10-30	11:00	Completed	Final report and analysis completed.
5	David Lee	202 Birch St	Seattle	WA	98101	206-555-7890	david.lee@example.com	2023-10-31	13:00	In Progress	Review of case file and follow-up actions.
6	Emily White	303 Cedar St	Portland	OR	97201	503-555-2345	emily.white@example.com	2023-11-01	15:00	Pending	Waiting for external agency response.
7	Michael Brown	404 Maple St	Denver	CO	80201	303-555-6789	michael.brown@example.com	2023-11-02	10:30	Completed	Case closed, all requirements met.
8	Sarah Davis	505 Spruce St	Phoenix	AZ	85001	602-555-0123	sarah.davis@example.com	2023-11-03	12:00	In Progress	Interview with subject and review of records.
9	James Wilson	606 Elm St	San Diego	CA	92101	619-555-4567	james.wilson@example.com	2023-11-04	14:00	Pending	Awaiting further information from the subject.
10	Lisa Anderson	707 Oak St	San Jose	CA	95101	408-555-8901	lisa.anderson@example.com	2023-11-05	11:30	Completed	Final report and analysis completed.
11	Christopher Taylor	808 Pine St	San Antonio	TX	78201	214-555-2345	christopher.taylor@example.com	2023-11-06	13:30	In Progress	Review of case file and follow-up actions.
12	Amanda Miller	909 Cedar St	San Jose	CA	95101	408-555-6789	amanda.miller@example.com	2023-11-07	15:30	Pending	Waiting for external agency response.
13	Kevin Hall	1010 Maple St	San Jose	CA	95101	408-555-0123	kevin.hall@example.com	2023-11-08	10:00	Completed	Case closed, all requirements met.
14	Nicole King	1111 Birch St	San Jose	CA	95101	408-555-4567	nicole.king@example.com	2023-11-09	12:30	In Progress	Interview with subject and review of records.
15	Brandon Scott	1212 Spruce St	San Jose	CA	95101	408-555-8901	brandon.scott@example.com	2023-11-10	14:30	Pending	Awaiting further information from the subject.
16	Stephanie Adams	1313 Elm St	San Jose	CA	95101	408-555-2345	stephanie.adams@example.com	2023-11-11	11:00	Completed	Final report and analysis completed.
17	Gregory Baker	1414 Oak St	San Jose	CA	95101	408-555-6789	gregory.baker@example.com	2023-11-12	13:00	In Progress	Review of case file and follow-up actions.
18	Heather Clark	1515 Pine St	San Jose	CA	95101	408-555-0123	heather.clark@example.com	2023-11-13	15:00	Pending	Waiting for external agency response.
19	Timothy Evans	1616 Cedar St	San Jose	CA	95101	408-555-4567	timothy.evans@example.com	2023-11-14	10:30	Completed	Case closed, all requirements met.
20	Christina Harris	1717 Maple St	San Jose	CA	95101	408-555-8901	christina.harris@example.com	2023-11-15	12:00	In Progress	Interview with subject and review of records.
21	Jonathan Lewis	1818 Birch St	San Jose	CA	95101	408-555-2345	jonathan.lewis@example.com	2023-11-16	14:00	Pending	Awaiting further information from the subject.
22	Karen Young	1919 Spruce St	San Jose	CA	95101	408-555-6789	karen.young@example.com	2023-11-17	11:30	Completed	Final report and analysis completed.
23	Benjamin Allen	2020 Elm St	San Jose	CA	95101	408-555-0123	benjamin.allen@example.com	2023-11-18	13:30	In Progress	Review of case file and follow-up actions.
24	Michelle Green	2121 Oak St	San Jose	CA	95101	408-555-4567	michelle.green@example.com	2023-11-19	15:30	Pending	Waiting for external agency response.
25	Christopher King	2222 Pine St	San Jose	CA	95101	408-555-8901	christopher.king@example.com	2023-11-20	10:00	Completed	Case closed, all requirements met.
26	Angela Wright	2323 Cedar St	San Jose	CA	95101	408-555-2345	angela.wright@example.com	2023-11-21	12:30	In Progress	Interview with subject and review of records.
27	Matthew Lopez	2424 Maple St	San Jose	CA	95101	408-555-6789	matthew.lopez@example.com	2023-11-22	14:30	Pending	Awaiting further information from the subject.
28	Rebecca Hill	2525 Birch St	San Jose	CA	95101	408-555-0123	rebecca.hill@example.com	2023-11-23	11:00	Completed	Final report and analysis completed.
29	Andrew Scott	2626 Spruce St	San Jose	CA	95101	408-555-4567	andrew.scott@example.com	2023-11-24	13:00	In Progress	Review of case file and follow-up actions.
30	Kimberly Adams	2727 Elm St	San Jose	CA	95101	408-555-8901	kimberly.adams@example.com	2023-11-25	15:00	Pending	Waiting for external agency response.
31	Joshua Baker	2828 Oak St	San Jose	CA	95101	408-555-2345	joshua.baker@example.com	2023-11-26	10:30	Completed	Case closed, all requirements met.
32	Brittany Clark	2929 Pine St	San Jose	CA	95101	408-555-6789	brittany.clark@example.com	2023-11-27	12:00	In Progress	Interview with subject and review of records.
33	Erica Evans	3030 Cedar St	San Jose	CA	95101	408-555-0123	erica.ev				



15. (continued)

The piston in the double-acting cylinder is required to outstroke **slowly** and **smoothly**. The speed of the instroke should not be changed.

(c) Describe how this would be achieved.

2

(d) State the type of **motion** a pneumatic piston produces when it outstrokes and then instrokes.

1

[END OF QUESTION PAPER]



ADDITIONAL SPACE FOR ANSWERS



ADDITIONAL SPACE FOR ANSWERS

Acknowledgement of copyright

- Question 5 Olivier Tabary/shutterstock.com
 Question 8 mapush/shutterstock.com
 Question 9 Editorial image: GiulianiBruno/shutterstock.com
 Question 10 Petr Malyshev/shutterstock.com
 Question 11 Photology1971/shutterstock.com
 Question 12 Editorial image: marekuliasz/shutterstock.com

